

**a Cross-kingdom
reference profiling**

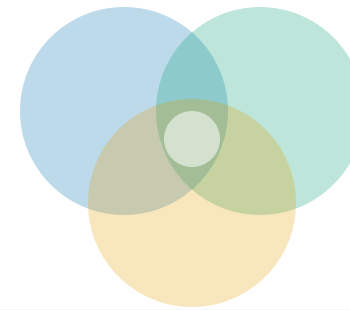


112 organisms · 6 kingdoms · 27 phyla

170 biological samples

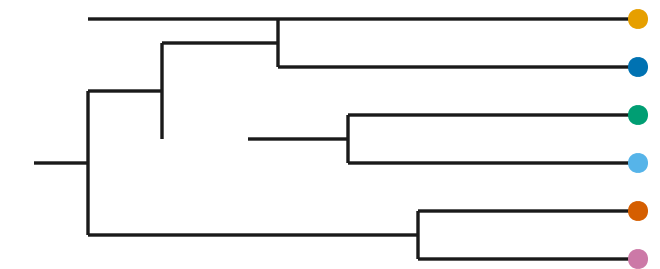
Untargeted LC–MS/MS
both polarities (ESI + / –)

**b Phylum-enriched
biomarker atlas**



cross-batch IndVal
+ composite scoring
enriched — not taxon-exclusive

**c Lipidomes recover
the tree of life**



UPGMA · Bray–Curtis
cross-batch features
chemotaxonomy ≈ phylogeny

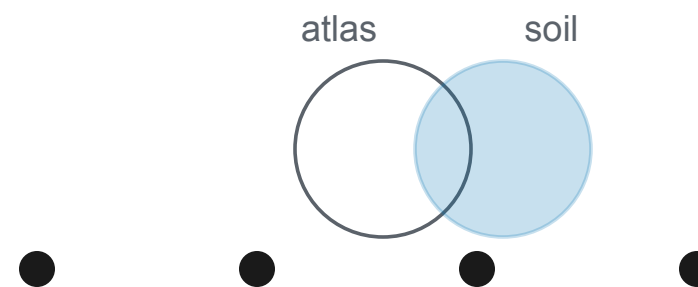
**f Six-kingdom community
decomposition**



ambient · drought · future climate · future + drought

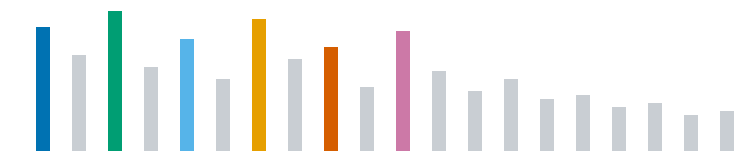
ClimGrass climate-manipulation
grassland soils
one extract per plot → six kingdoms

**e Fingerprint matching
& correction**



partial fingerprint matching
+ 4-step quantification correction
fold-change-weighted Bray–Curtis

**d Distributed
fingerprints (SIMPER)**



SIMPER extracts the features
that separate each phylum
signal is distributed, not single markers